



LIBRARY MANAGEMENT SYSTEM

Your next-door library



MARCH 7, 2025
LIBRARY



What we do

At our library, we want to help members find the perfect book for their next read. Not just any book, but exceptional books filled with inspiring and life-enriching stories. We would like to create an easy-to-use platform that allows our members to discover books and track their borrowing history.

As the manager, I face daily challenges in keeping up with the needs of our more than 1000 members. Our current system is outdated and relies on manual processes for tracking books, member records, and borrowing transactions. This system leads to inefficiencies, increased errors, and delays in responding to member requests.

To solve these issues, we are proposing a new Library Management System. This system will bring everything together in one place—tracking books (both for borrowing and sales), authors, member details, and borrow records in real time. It will simplify our operations, reduce mistakes, and allow us to respond quickly to our members' needs.

Our goal is to make the library experience comfortable and enjoyable for everyone. This solution directly addresses our current challenges and is a crucial step toward modernizing our library services for a smoother, more efficient experience.

Business and Mission

- Our mission of our library is to provide personalized service to all our members. We want to create a user-friendly, innovative, and accessible for our members to discover, borrow, buy and enjoy books.
- The library has been growing fast, and as part of our plan, we liked to improve the system for managing member data, books, and borrowing records. The current method of data management needs to be improved by storing it in a real-time, more accessible, and readily updateable database.
- In addition the system will allow members to access their past records to provide personalized service to any member.

Information Requirements

The Library Management System will need to handle information, such as:

- Member Information: First and last names, phone numbers, email addresses, Home addresses, and membership details.
- Book Information: Including the book's title, author, genre, publish year, and availability.
- Borrowing Information: Keeping track of when a book was borrowed, due dates, overdue status, and past borrowing history.
- Statistic of bought books: Trends are coming fast, and demands are high. Statistic of bought books should be tracked to not be out of stock wenn demands are follwing in.
- Book Preferences: Members will have the option to specify preferred genres or authors, so the librarians can provide personalized book recommendations simply by accessing the member records.
- Librarian Interaction: Librarians will need access to member records, the ability to track borrowed or bought books.

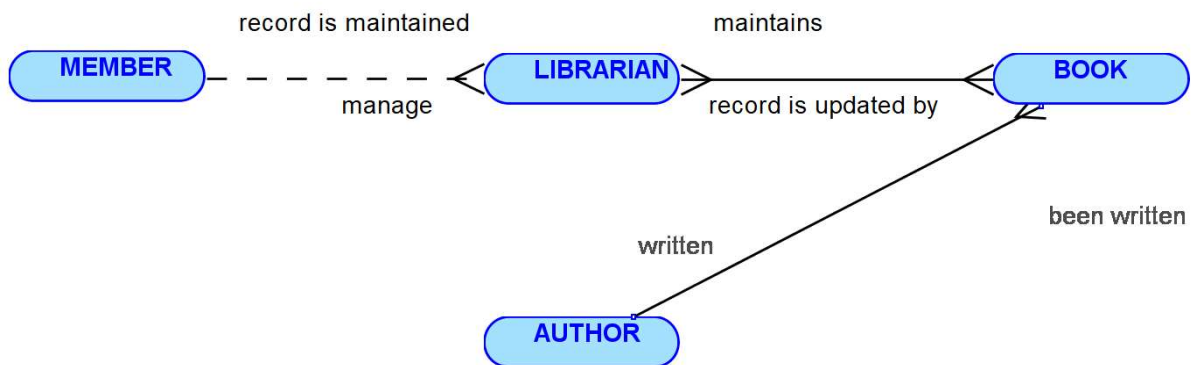
Business Rules

- Personalized Service: Each member should be able to access their borrowing or bought history.
- Book Borrowing: A member can borrow a book, but if the book is already borrowed by someone else, they can request a reservation. When the book becomes available, the member will be notified.
- Book ordering: A member can order a book if isn't avaiable in the library. When the book arrives, the member will be notified. The member has to pay half of the orders book as a garantie.
- Librarian Management: Librarians update the database. They need to add new members, new books, and update the status of borrowed and bought books.

Assumptions

- Data Privacy: Members are aware that their borrowing history and preferences of genres or authors will be stored and accessed by librarians, and they are agreeable to this.
- Book Reservations: Members can reserve books that are currently not available and will be notified once the books are available for borrowing.
- Book order: Members can order books that are currently not available in the library will be notified once the books have arrived.

Conceptual model (MCD)



Entity Relationship Diagram (MELD)

- Write the sentences for each relationship, in both directions

Each Preference may belong to one and only one member. Each Member may enter one or more Preference.

Each Member may can request one or more Reservation. Each Reservation must be made by one and only one Member.

Each Member may buy one or more Purchase. Each Purchase must be record by one and only one Member.

Each Member may place one or more Book Order. Each Book Order must be placed by one and only one Member.

Each Member may have it's record maintained by one or more Librarian. Each Librarian may manage by one and only one Member.

Each Member may create one or more borrowing. Each borrowing must belong to one and only one Member.

Each Book may can have one or more Reservation. Each Reservation must be recorded by one and only one Book.

Each Book may can be purchased by one or more Purchase. Each Purchase must be recorded by one and only one Book.

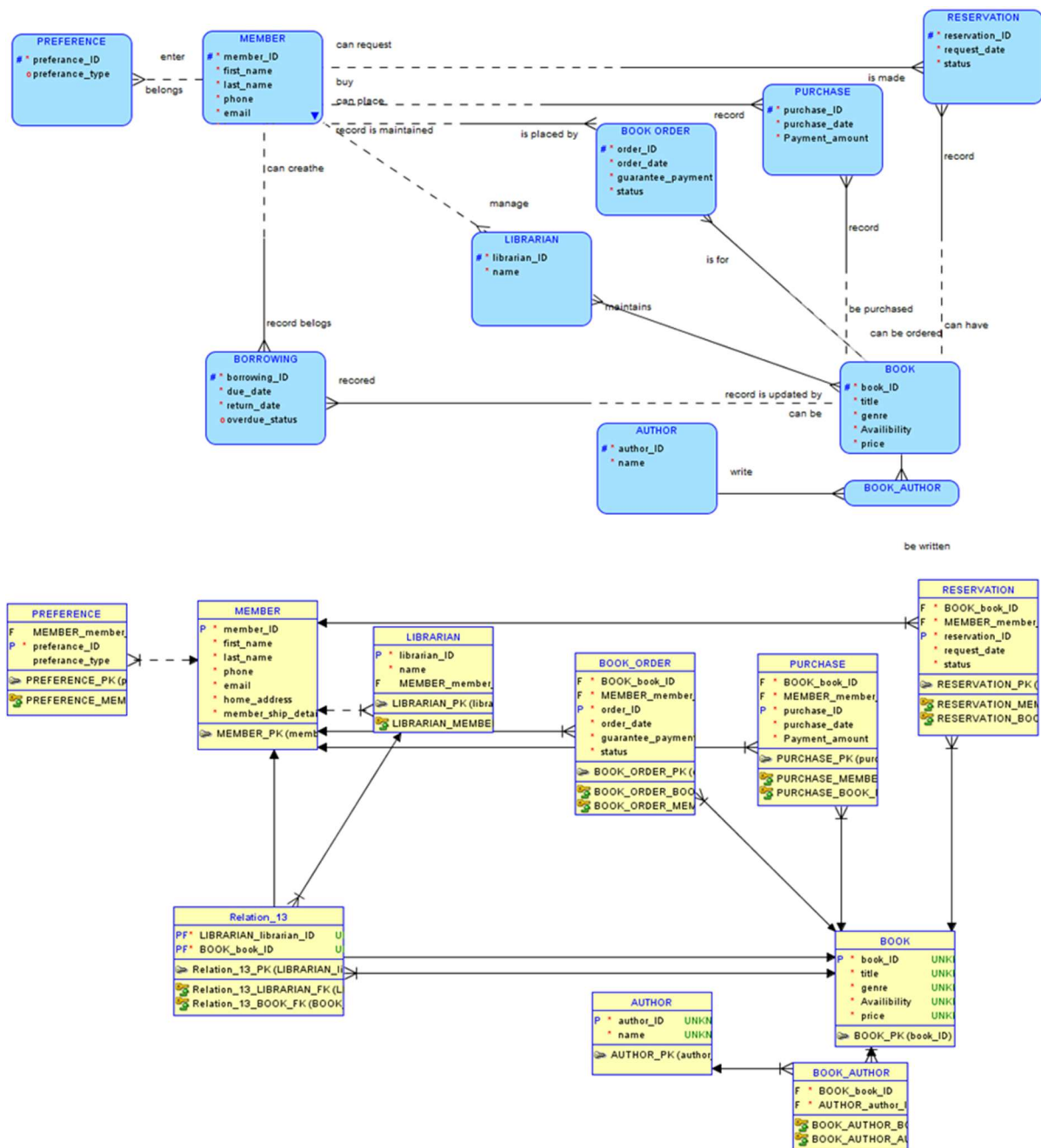
Each Book may can be ordered by one or more Book Order. Each Book Order must be for one and only one Book.

Each Book may can have it's record updated by one or more borrowing. Each borrowing must be recorded for one and only one Book.

Each Book must have it's record updated by one or more Librarian. Each Librarian must maintain one or more Book.

Each Book must be written by one or more Author. Each Author must write one or more Book.

- Map the relationships.



- Draw the matrix diagram

	Member	Preference	Reservation	Book Order	borrowing	Librarian	Author	Book	Purchase
Member		enter	request	place	create	record			buy
Preference	belong								
Reservation	made							recorded	
Book Order	placed							must be	
borrowing	belong							recorded	
Librarian	manage								maintain
Author								write	
Book			have	ordered	record update	record updated	written		purchased
Purchase	recorded							recorded	

MAPPING RELATIONSHIPS		
Preference		
Key-Types	Optimality	Column-name
PK	#	Preference_ID
	o	Preference_type
FK	#	Member_ID
Member		
Key-Types	Optimality	Column-name
PK	#	Member_ID
	*	first_name
	*	last_name
	*	phone
	*	email
	*	home_address
	*	member_ship_data
Librarian		
Key-Types	Optimality	Column-name
PK(1)	#	Librarian_ID
PK(12)	#	Member_ID
	*	
Book		
Key-Types	Optimality	Column-name
PK(1)	#	Book_ID
PK(12)	#	title
	*	genre
		Availibility
		Price
Book Order		
Key-Types	Optimality	Column-name
PK	#	Book_Order_ID
FK	#	Member_ID
FK	#	Book_ID
	*	order_date
	*	garantee_payment
	*	home_address
	*	member_ship_data
Purchase		
Key-Types	Optimality	Column-name
FK	#	Book_Order_ID
FK	#	Member_ID
PK	#	Purchase_ID
	*	Purchase_date
	*	payment_amout
Reservation		
Key-Types	Optimality	Column-name
FK	#	Book_Order_ID
FK	#	Member_ID
PK	#	Reservation_ID
	*	request_date
	*	status
FK	#	Author_ID
Author		
Key-Types	Optimality	Column-name
PK(1)	#	Author_ID
PK(12)	#	Book_id
		name

Create Tables:

- Write the commands to create each table
- Add constraints: write the commands to add PK and FK

```
CREATE TABLE Library_Member(
  member_ID number(6) CONSTRAINT lib_mem_id_pk PRIMARY KEY,
  first_name varchar2(40) NOT NULL,
  last_name varchar2(40) NOT NULL,
  phone varchar2(40),
  email varchar2(40),
  home_address varchar2(60) NOT NULL,
  member_ship_detail varchar2(40),
  CONSTRAINT lib_phone_email_uk UNIQUE (phone, email)
);
```

Column Name	Data Type	Nullable	Default	Primary Key
MEMBER_ID	NUMBER(6,0)	No	-	1
FIRST_NAME	VARCHAR2(40)	No	-	-
LAST_NAME	VARCHAR2(40)	No	-	-
PHONE	VARCHAR2(40)	Yes	-	-
EMAIL	VARCHAR2(40)	Yes	-	-
HOME_ADDRESS	VARCHAR2(60)	No	-	-
MEMBER_SHIP_DETAIL	VARCHAR2(40)	Yes	-	-

```
CREATE TABLE Library_Book(
  book_ID number(6) CONSTRAINT lib_book_id_pk PRIMARY KEY,
  title varchar2(40) NOT NULL,
  genre varchar2(40),
  availability varchar2(40),
  price varchar2(40) NOT NULL
);
```

Column Name	Data Type	Nullable	Default	Primary Key
BOOK_ID	NUMBER(6,0)	No	-	1
TITLE	VARCHAR2(40)	No	-	-
GENRE	VARCHAR2(40)	Yes	-	-
AVAILABILITY	VARCHAR2(40)	Yes	-	-
PRICE	VARCHAR2(40)	No	-	-

```
CREATE TABLE Library_Preference(
  preference_ID number(6) CONSTRAINT lib_pref_id_pk PRIMARY KEY,
  preference_type varchar2(40)
);
```

Column Name	Data Type	Nullable	Default	Primary Key
PREFERENCE_ID	NUMBER(6,0)	No	-	1
PREFERENCE_TYPE	VARCHAR2(40)	Yes	-	-


```
CREATE TABLE Library_Librarian(
  librarian_ID number(6) CONSTRAINT lib_lib_id_pk PRIMARY KEY,
  name varchar2(40),
  member_ID number(6),
  CONSTRAINT lib_mem_id_fk FOREIGN KEY (member_ID) REFERENCES Library_Member(member_ID)
);
```

Column Name	Data Type	Nullable	Default	Primary Key
LIBRARIAN_ID	NUMBER(6,0)	No	-	1
NAME	VARCHAR2(40)	Yes	-	-
MEMBER_ID	NUMBER(6,0)	Yes	-	-

```
CREATE TABLE Library_Book_Order(
  book_order_ID number(6) CONSTRAINT lib_book_ord_id_pk PRIMARY KEY,
  order_date varchar2(40),
  guarantee_payment varchar2(40),
  status varchar2(40),
  member_ID number(6),
  book_ID number(6),
  CONSTRAINT lib1_mem_id_fk FOREIGN KEY (member_ID) REFERENCES Library_Member(member_ID),
  CONSTRAINT lib1_book_id_fk FOREIGN KEY (book_ID) REFERENCES Library_Book(book_ID)
);
```

Column Name	Data Type	Nullable	Default	Primary Key
BOOK_ORDER_ID	NUMBER(6,0)	No	-	1
ORDER_DATE	VARCHAR2(40)	Yes	-	-
GARANTEE_PAYMENT	VARCHAR2(40)	Yes	-	-
STATUS	VARCHAR2(40)	Yes	-	-
MEMBER_ID	NUMBER(6,0)	Yes	-	-
BOOK_ID	NUMBER(6,0)	Yes	-	-

```
CREATE TABLE Library_Purchase(
  purchase_ID number(6) CONSTRAINT lib_pur_id_pk PRIMARY KEY,
  purchase_date varchar2(40),
  payment_amount varchar2(40) NOT NULL,
  member_ID number(6),
  book_ID number(6),
  CONSTRAINT lib2_mem_id_fk FOREIGN KEY (member_ID) REFERENCES Library_Member(member_ID),
  CONSTRAINT lib2_book_id_fk FOREIGN KEY (book_ID) REFERENCES Library_Book(book_ID)
);
```

Column Name	Data Type	Nullable	Default	Primary Key
PURCHASE_ID	NUMBER(6,0)	No	-	1
PURCHASE_DATE	VARCHAR2(40)	Yes	-	-
PAYMENT_AMOUNT	VARCHAR2(40)	No	-	-
MEMBER_ID	NUMBER(6,0)	Yes	-	-
BOOK_ID	NUMBER(6,0)	Yes	-	-

```
CREATE TABLE Library_Borrowing(
  borrowing_ID number(6) CONSTRAINT lib_borr_id_pk PRIMARY KEY,
  due_date varchar2(40),
  return_date varchar2(40),
  overdue_status varchar2(40),
  member_ID number(6),
  book_ID number(6),
  CONSTRAINT lib3_mem_id_fk FOREIGN KEY (member_ID) REFERENCES Library_Member(member_ID),
  CONSTRAINT lib3_book_id_fk FOREIGN KEY (book_ID) REFERENCES Library_Book(book_ID)
);
```

Column Name	Data Type	Nullable	Default	Primary Key
BORROWING_ID	NUMBER(6,0)	No	-	1
DUE_DATE	VARCHAR2(40)	Yes	-	-
RETURN_DATE	VARCHAR2(40)	Yes	-	-
OVERDUE_STATUS	VARCHAR2(40)	Yes	-	-
MEMBER_ID	NUMBER(6,0)	Yes	-	-
BOOK_ID	NUMBER(6,0)	Yes	-	-

```
CREATE TABLE Library_Reservation(
  reservation_ID number(6) CONSTRAINT lib_res_id_pk PRIMARY KEY,
  request_date varchar2(40),
  status varchar2(40),
  member_ID number(6),
  book_ID number(6),
  CONSTRAINT lib4_mem_id_fk FOREIGN KEY (member_ID) REFERENCES Library_Member(member_ID),
  CONSTRAINT lib4_book_id_fk FOREIGN KEY (book_ID) REFERENCES Library_Book(book_ID)
);
```

Column Name	Data Type	Nullable	Default	Primary Key
RESERVATION_ID	NUMBER(6,0)	No	-	1
REQUEST_DATE	VARCHAR2(40)	Yes	-	-
STATUS	VARCHAR2(40)	Yes	-	-
MEMBER_ID	NUMBER(6,0)	Yes	-	-
BOOK_ID	NUMBER(6,0)	Yes	-	-

```
CREATE TABLE Library_Author(
  author_ID number(6) CONSTRAINT lib_aut_id_pk PRIMARY KEY,
  name varchar2(40)
);
```

Column Name	Data Type	Nullable	Default	Primary Key
AUTHOR_ID	NUMBER(6,0)	No	-	1
NAME	VARCHAR2(40)	Yes	-	-

```
CREATE TABLE Library_Book_Author(
  author_ID number(6),
  book_ID number(6),
  CONSTRAINT lib_aut_id_fk FOREIGN KEY (author_ID) REFERENCES Library_Author(author_ID),
  CONSTRAINT lib5_book_id_fk FOREIGN KEY (book_ID) REFERENCES Library_Book(book_ID)
);
```

Column Name	Data Type	Nullable	Default	Primary Key
AUTHOR_ID	NUMBER(6,0)	Yes	-	-
BOOK_ID	NUMBER(6,0)	Yes	-	-

Add data to tables write the commands to insert records

```

insert all into Library_Member
values (101, 'Emma', 'Stone', '1234567890', 'emma.stone@example.com', '123 Oak Street',
'Gold Member')
INTO Library_Member VALUES (102, 'Liam', 'Johnson', '2345678901',
'liam.johnson@example.com', '456 Pine Avenue', 'Silver Member')
INTO Library_Member VALUES (103, 'Olivia', 'Williams', '3456789012',
'olivia.williams@example.com', '789 Maple Blvd', 'Platinum Member')
INTO Library_Member VALUES (104, 'Noah', 'Brown', '4567890123',
'noah.brown@example.com', '321 Cedar Lane', 'Silver Member')
INTO Library_Member VALUES (105, 'Ava', 'Davis', '5678901234', 'ava.davis@example.com',
'654 Birch Way', 'Gold Member')
INTO Library_Member VALUES (106, 'Oliver', 'Miller', '3849201756',
'oliver.miller@example.com', '123 Silverbrook Avenue', 'Gold Member')
INTO Library_Member VALUES (107, 'William', 'Smith', '7291048365',
'william.smith@example.com', '456 Hawthorn Crescent', 'Silver Member')
INTO Library_Member VALUES (108, 'Grace', 'Li', '1057384920', 'grace.li.i@example.com', '789
Bluebell Hill Street', 'Platinum Member')
INTO Library_Member VALUES (109, 'Amelia', 'Garcia', '6482910375',
'amelia.garcia@example.com', '321 Maple Hollow Lane', 'Silver Member')
INTO Library_Member VALUES (110, 'Mia', 'Levi', '5928374610', 'mia.levi@example.com', '654
Maple Hollow Lane', 'Gold Member')
select 1 from dual;
  
```

MEMBER_ID	FIRST_NAME	LAST_NAME	PHONE	EMAIL	HOME_ADDRESS	MEMBER_SHIP_DETAIL
106	Oliver	Miller	3849201756	oliver.miller@example.com	123 Silverbrook Avenue	Gold Member
107	William	Smith	7291048365	william.smith@example.com	456 Hawthorn Crescent	Silver Member
108	Grace	Li	1057384920	grace.li.i@example.com	789 Bluebell Hill Street	Platinum Member
109	Amelia	Garcia	6482910375	amelia.garcia@example.com	321 Maple Hollow Lane	Silver Member
110	Mia	Levi	5928374610	mia.levi@example.com	654 Maple Hollow Lane	Gold Member
101	Emma	Stone	1234567890	emma.stone@example.com	123 Oak Street	Gold Member
102	Liam	Johnson	2345678901	liam.johnson@example.com	456 Pine Avenue	Silver Member
103	Olivia	Williams	3456789012	olivia.williams@example.com	789 Maple Blvd	Platinum Member
104	Noah	Brown	4567890123	noah.brown@example.com	321 Cedar Lane	Silver Member
105	Ava	Davis	5678901234	ava.davis@example.com	654 Birch Way	Gold Member

insert all into Library_Book

values (201, 'A Winter's Promise', 'Fantasy', 'Available', '15.99')

INTO Library_Book VALUES (202, 'Absolu', 'Romance', 'Available', '12.50')

INTO Library_Book VALUES (203, 'Scythe', 'Dystopian', 'Unavailable', '18.00')

INTO Library_Book VALUES (204, 'Divergent', 'Science Fiction', 'Available', '14.25')

INTO Library_Book VALUES (205, 'Harry Potter', 'Fantasy', 'Available', '22.00')

INTO Library_Book VALUES (206, 'The Mirror Visitor', 'Fantasy', 'Available', '16.75')

INTO Library_Book VALUES (207, 'The Inheritance Games', 'Mystery', 'Available', '13.40')

INTO Library_Book VALUES (208, 'Renegades', 'Superhero', 'Unavailable', '17.60')

INTO Library_Book VALUES (209, 'The Night Circus', 'Fantasy', 'Available', '15.20')

INTO Library_Book VALUES (210, 'The Maze Runner', 'Dystopian', 'Available', '14.80')

select 1 from dual;

BOOK_ID	TITLE	GENRE	AVAILABILITY	PRICE
206	The Mirror Visitor	Fantasy	Available	16.75
207	The Inheritance Games	Mystery	Available	13.40
208	Renegades	Superhero	Unavailable	17.60
209	The Night Circus	Fantasy	Available	15.20
210	The Maze Runner	Dystopian	Available	14.80
201	A Winter's Promise	Fantasy	Available	15.99
202	Absolu	Romance	Available	12.50
203	Scythe	Dystopian	Unavailable	18.00
204	Divergent	Science Fiction	Available	14.25
205	Harry Potter	Fantasy	Available	22.00

insert all into Library_Preference

values (301, 'Fantasy')

INTO Library_Preference VALUES (302, 'Romance')

INTO Library_Preference VALUES (303, 'Dystopian')

INTO Library_Preference VALUES (304, 'Science Fiction')

INTO Library_Preference VALUES (305, 'Adventure')

INTO Library_Preference VALUES (306, 'Mystery')

INTO Library_Preference VALUES (307, 'Historical Fiction')

INTO Library_Preference VALUES (308, 'Thriller')

INTO Library_Preference VALUES (309, 'Superhero')

INTO Library_Preference VALUES (310, 'Horror')

select 1 from dual;

PREFERENCE_ID	PREFERENCE_TYPE
301	Fantasy
302	Romance
303	Dystopian
304	Science Fiction
305	Adventure
306	Mystery
307	Historical Fiction
308	Thriller
309	Superhero
310	Horror

```

insert all into Library_Librarian
values (401, 'Sophia Clark', 101)
INTO Library_Librarian VALUES (402, 'Jackson Miller', 102)
INTO Library_Librarian VALUES (403, 'Isabella Wilson', 103)
INTO Library_Librarian VALUES (404, 'Lucas Moore', 104)
INTO Library_Librarian VALUES (405, 'Mia Taylor', 105)
INTO Library_Librarian VALUES (406, 'Ethan Johnson', 106)
INTO Library_Librarian VALUES (407, 'Ava Martinez', 107)
INTO Library_Librarian VALUES (408, 'Noah Anderson', 108)
INTO Library_Librarian VALUES (409, 'Emma Thompson', 109)
INTO Library_Librarian VALUES (410, 'Liam White', 110)
select 1 from dual;

```

LIBRARIAN_ID	NAME	MEMBER_ID
401	Sophia Clark	101
402	Jackson Miller	102
403	Isabella Wilson	103
404	Lucas Moore	104
405	Mia Taylor	105
406	Ethan Johnson	106
407	Ava Martinez	107
408	Noah Anderson	108
409	Emma Thompson	109
410	Liam White	110

```

insert all into Library_Book_Order
values (501, '1-Jan-2024', 'Yes', 'Pending', 101, 201)
INTO Library_Book_Order VALUES (502, '1-Jan-2024', 'No', 'Pending', 102, 202)
INTO Library_Book_Order VALUES (503, '1-Jan-2024', 'Yes', 'Delivered', 103, 203)
INTO Library_Book_Order VALUES (504, '1-Jan-2024', 'Yes', 'Pending', 104, 204)
INTO Library_Book_Order VALUES (505, '1-Jan-2024', 'No', 'Delivered', 105, 205)
INTO Library_Book_Order VALUES (506, '2-Jan-2024', 'Yes', 'Pending', 106, 206)
INTO Library_Book_Order VALUES (507, '2-Jan-2024', 'No', 'Delivered', 107, 207)
INTO Library_Book_Order VALUES (508, '2-Jan-2024', 'Yes', 'Delivered', 108, 208)
INTO Library_Book_Order VALUES (509, '2-Jan-2024', 'No', 'Pending', 109, 209)
INTO Library_Book_Order VALUES (510, '2-Jan-2024', 'Yes', 'Delivered', 110, 210)
select 1 from dual;

```

BOOK_ORDER_ID	ORDER_DATE	GUARANTEE_PAYMENT	STATUS	MEMBER_ID	BOOK_ID
506	2-Jan-2024	Yes	Pending	106	206
507	2-Jan-2024	No	Delivered	107	207
508	2-Jan-2024	Yes	Delivered	108	208
509	2-Jan-2024	No	Pending	109	209
510	2-Jan-2024	Yes	Delivered	110	210
501	1-Jan-2024	Yes	Pending	101	201
502	1-Jan-2024	No	Pending	102	202
503	1-Jan-2024	Yes	Delivered	103	203
504	1-Jan-2024	Yes	Pending	104	204
505	1-Jan-2024	No	Delivered	105	205

insert all into Library_Purchase

values (601, '1-Jan-2024', '15.99', 101, 201)

INTO Library_Purchase VALUES (602, '1-Jan-2024', '12.50', 102, 202)

INTO Library_Purchase VALUES (603, '1-Jan-2024', '18.00', 103, 203)

INTO Library_Purchase VALUES (604, '1-Jan-2024', '14.25', 104, 204)

INTO Library_Purchase VALUES (605, '1-Jan-2024', '22.00', 105, 205)

INTO Library_Purchase VALUES (606, '2-Jan-2024', '16.75', 106, 206)

INTO Library_Purchase VALUES (607, '2-Jan-2024', '13.40', 107, 207)

INTO Library_Purchase VALUES (608, '2-Jan-2024', '17.60', 108, 208)

INTO Library_Purchase VALUES (609, '2-Jan-2024', '15.20', 109, 209)

INTO Library_Purchase VALUES (610, '2-Jan-2024', '14.80', 110, 210)

select 1 from dual;

PURCHASE_ID	PURCHASE_DATE	PAYMENT_AMOUNT	MEMBER_ID	BOOK_ID
601	1-Jan-2024	15.99	101	201
602	1-Jan-2024	12.50	102	202
603	1-Jan-2024	18.00	103	203
604	1-Jan-2024	14.25	104	204
605	1-Jan-2024	22.00	105	205
606	2-Jan-2024	16.75	106	206
607	2-Jan-2024	13.40	107	207
608	2-Jan-2024	17.60	108	208
609	2-Jan-2024	15.20	109	209
610	2-Jan-2024	14.80	110	210

insert all into Library_Borrowing

values (705, '1-Jan-2024', '1-Jan-2024', 'No', 105, 205)

INTO Library_Borrowing VALUES (701, '1-Jan-2024', '1-Jan-2024', 'No', 101, 201)

INTO Library_Borrowing VALUES (702, '1-Jan-2024', '1-Jan-2024', 'No', 102, 202)

INTO Library_Borrowing VALUES (703, '1-Jan-2024', '1-Jan-2024', 'No', 103, 203)

INTO Library_Borrowing VALUES (704, '1-Jan-2024', '1-Jan-2024', 'No', 104, 204)

INTO Library_Borrowing VALUES (706, '2-Jan-2024', '2-Jan-2024', 'No', 106, 206)

INTO Library_Borrowing VALUES (707, '2-Jan-2024', '2-Jan-2024', 'No', 107, 207)

INTO Library_Borrowing VALUES (708, '2-Jan-2024', '2-Jan-2024', 'No', 108, 208)

INTO Library_Borrowing VALUES (709, '2-Jan-2024', '2-Jan-2024', 'No', 109, 209)

INTO Library_Borrowing VALUES (710, '2-Jan-2024', '2-Jan-2024', 'No', 110, 210)

select 1 from dual;

BORROWING_ID	DUE_DATE	RETURN_DATE	OVERDUE_STATUS	MEMBER_ID	BOOK_ID
706	2-Jan-2024	2-Jan-2024	No	106	206
707	2-Jan-2024	2-Jan-2024	No	107	207
708	2-Jan-2024	2-Jan-2024	No	108	208
709	2-Jan-2024	2-Jan-2024	No	109	209
710	2-Jan-2024	2-Jan-2024	No	110	210
705	1-Jan-2024	1-Jan-2024	No	105	205
701	1-Jan-2024	1-Jan-2024	No	101	201
702	1-Jan-2024	1-Jan-2024	No	102	202
703	1-Jan-2024	1-Jan-2024	No	103	203
704	1-Jan-2024	1-Jan-2024	No	104	204

insert all into Library_Reservation

values (801, '1-Jan-2024', 'Pending', 101, 201)

INTO Library_Reservation VALUES (802, '1-Jan-2024', 'Confirmed', 102, 202)

INTO Library_Reservation VALUES (803, '1-Jan-2024', 'Pending', 103, 203)

INTO Library_Reservation VALUES (804, '1-Jan-2024', 'Confirmed', 104, 204)

INTO Library_Reservation VALUES (805, '1-Jan-2024', 'Pending', 105, 205)

INTO Library_Reservation VALUES (806, '2-Jan-2024', 'Confirmed', 106, 206)

INTO Library_Reservation VALUES (807, '2-Jan-2024', 'Pending', 107, 207)

INTO Library_Reservation VALUES (808, '2-Jan-2024', 'Confirmed', 108, 208)

INTO Library_Reservation VALUES (809, '2-Jan-2024', 'Pending', 109, 209)

INTO Library_Reservation VALUES (810, '2-Jan-2024', 'Confirmed', 110, 210)

select 1 from dual;

RESERVATION_ID	REQUEST_DATE	STATUS	MEMBER_ID	BOOK_ID
806	2-Jan-2024	Confirmed	106	206
807	2-Jan-2024	Pending	107	207
808	2-Jan-2024	Confirmed	108	208
809	2-Jan-2024	Pending	109	209
810	2-Jan-2024	Confirmed	110	210
801	1-Jan-2024	Pending	101	201
802	1-Jan-2024	Confirmed	102	202
803	1-Jan-2024	Pending	103	203
804	1-Jan-2024	Confirmed	104	204
805	1-Jan-2024	Pending	105	205

insert all into Library_Author

values (901, 'Christelle Dabos')

INTO Library_Author VALUES (902, 'Margot Dessenne')

INTO Library_Author VALUES (903, 'Neal Shusterman')

INTO Library_Author VALUES (904, 'Veronica Roth')

INTO Library_Author VALUES (905, 'J.K. Rowling')

INTO Library_Author VALUES (906, 'Christelle Dabos')

INTO Library_Author VALUES (907, 'Jennifer Lynn Barnes')

INTO Library_Author VALUES (908, 'Marissa Meyer')

INTO Library_Author VALUES (909, 'Erin Morgenstern')

INTO Library_Author VALUES (910, 'James Dashner')

select 1 from dual;

AUTHOR_ID	NAME
906	Christelle Dabos
907	Jennifer Lynn Barnes
908	Marissa Meyer
909	Erin Morgenstern
910	James Dashner
901	Christelle Dabos
902	Margot Dessenne
903	Neal Shusterman
904	Veronica Roth
905	J.K. Rowling

```
insert all into Library_Book_Author
values (901, 201)
INTO Library_Book_Author VALUES (902, 202)
INTO Library_Book_Author VALUES (903, 203)
INTO Library_Book_Author VALUES (904, 204)
INTO Library_Book_Author VALUES (905, 205)
INTO Library_Book_Author VALUES (906, 206)
INTO Library_Book_Author VALUES (907, 207)
INTO Library_Book_Author VALUES (908, 208)
INTO Library_Book_Author VALUES (909, 209)
INTO Library_Book_Author VALUES (910, 210)
select 1 from dual;
```

AUTHOR_ID	BOOK_ID
901	201
902	202
903	203
904	204
905	205
906	206
907	207
908	208
909	209
910	210

Create forms:

- Form

	Title ↑	Genre	Availability	Price
	A Winter's Promise	Fantasy	Available	15.99
	Absolu	Romance	Available	12.50
	Divergent	Science Fiction	Available	14.25
	Harry Potter	Fantasy	Available	22.00
	Scythe	Dystopian	Unavailable	18.00

Purpose: Used to add or edit book details.

Description: This form ensures accurate data entry for each book, including fields like title, genre, availability, and price.

- Interactive grid

☐	Title	Genre	Availability	Price
☒	A Winter's Promise	Fantasy	Available	15.99
☐	Absolu	Romance	Available	12.50
☐	Scythe	Dystopian	Unavailable	18.00
☐	Divergent	Science Fiction	Available	14.25
☐	Harry Potter	Fantasy	Available	22.00

Purpose: Manage multiple book records at once.

Description: A spreadsheet-style view that allows inline editing, sorting, and filtering—ideal for quick updates

- Interactive report

	Title ↑	Genre	Availability	Price
	A Winter's Promise	Fantasy	Available	15.99
	Absolu	Romance	Available	12.50
	Divergent	Science Fiction	Available	14.25
	Harry Potter	Fantasy	Available	22.00
	Scythe	Dystopian	Unavailable	18.00

Purpose: View and analyze book data.

Description: A dynamic report that displays book records with filtering options—great for tracking availability or generating insights.